Karthikeya Vellanki

650-942-5065 | vellankikarthikeya@gmail.com | linkedin.com/in/karthikeya-vellanki | github.com/KarthikeyaVell

Education

University of Texas at Dallas Richardson, TX

B.S. in Computer Information Systems and Technology *May 2028*

- Relevant Coursework: Networks & Operating Systems, Database Fundamentals, Network Security, Data Governance, Risk Analytics

Certifications

- Cisco Certified Network Associate | ISC2 Certified in Cybersecurity (CC) | Google Cybersecurity Certificate — Linux, SQL, Security Fundamentals | Java Information Technology Specialist

Technical Experience

SOC Home Lab Infrastructure June 2026 – Present

Self-Directed Project *Remote*

- Deployed Proxmox VE hypervisor to host and manage multiple virtualized services (OPNsense firewall, Pi-hole DNS sinkhole, Wazuh SIEM) via SSH, simulating enterprise-level infrastructure on commodity hardware
- Configured OPNsense as a virtualized dual-WAN firewall (IPv4/IPv6) implementing WAN/LAN network segmentation, stateful firewall rules governing TCP/UDP traffic, NAT, and Suricata-based intrusion detection, sustaining 5+ consecutive days of uptime with zero unplanned downtime
- Deployed WireGuard VPN (public/private key authentication, peer configuration, port-forwarding on an AT&T gateway) for secure remote lab access with zero exposed public-facing ports, and set up Pi-hole as a network-wide DNS sinkhole filtering traffic against 99,000+ blocklisted malicious/telemetry domains across 4 active clients
- Deployed Wazuh SIEM/XDR to monitor 2 active endpoint agents, triaging 360+ security alerts (72 medium, 288 low severity) within a 24-hour window with zero critical or high-severity incidents

Projects

Secure Multi-Floor Campus Network | *Cisco Packet Tracer* January 2026

- Designed a scalable, hierarchical campus network (core, distribution, access layers) spanning 4 buildings with 4 floors each, segmented into 6 departmental VLANs to support end-user connectivity and operational efficiency across a simulated enterprise
- Architected a separate, isolated network branch hosting a shared staff-and-student-lab server, logically and physically segmented from the main departmental network for centralized resource access
- Implemented inter-VLAN routing and configured DHCP/RIP protocols for dynamic IP addressing and inter-network communication across all building and department subnets
- Tested and validated network performance, confirming 100% inter-VLAN connectivity with sub-3ms ping latency across the simulated multi-building enterprise infrastructure

Multilingual AI Financial Assistant | *TypeScript, Node.js, OpenAI GPT-4o mini, ElevenLabs* December 2025

- Built a multilingual, AI-powered financial literacy web app in UTDHacks '25, that lets users view and analyze transactions and interact with a Generative AI assistant through text and voice
- Integrated LLM APIs (OpenAI GPT-4o mini, Google Gemini) and ElevenLabs speech synthesis to power a conversational chatbot delivering personalized financial recommendations
- Developed a custom financial model and interactive learning modules to generate tailored guidance and improve users' financial literacy

Technical Skills

Networking: OSI Layers, VLANs, Inter-VLAN Routing, 802.1Q Trunking, Router-on-a-Stick, Subnetting, DHCP, DNS, Cisco Packet Tracer, TCP/IP, NAT, RIP, SSH
Systems, Virtualization & Cloud: Proxmox VE, LXC Containers, VM Deployment & Management, Linux, FreeBSD, Bash, Windows Server - Active Directory
Server & Database Administration: SQL, Linux/Windows Server Administration, Service Deployment & Monitoring, Firewall Configuration (OPNsense, pfSense)
Security: Network Segmentation, Port Security, VPN Tunneling (WireGuard, Tailscale), SIEM/XDR (Wazuh), Intrusion Detection (Suricata)
Programming: Python, SQL, Java, C++, JavaScript, TypeScript, HTML/CSS